

SAKSHAM JAIN

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EDUCATION

SEP 2022 - Present	Ph.D., Statistics - UNIVERSITY OF WASHINGTON, SEATTLE Advisors: Alex Luedtke , Ting Ye
AUG 2020 - MAY 2022	M.S., ECE (Machine Learning) - DUKE UNIVERSITY Advisors: Cynthia Rudin , Alex Volfovsky , Sudeepa Roy Grade: 3.9/4
AUG 2015 - JUN 2019	B.E., (Robotics & Automation) - NSIT, UNIVERSITY OF DELHI Advisor: Umang Soni Grade: 7.7/10

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

Jain, S., Ye, T. (2026). Longitudinal Causal Inference with Multiple Continuous Exposures: An Application to Proteomics. *In Preparation*.

Jain, S., Luedtke, A. (2026). Conditional Distributional Treatment Effects: Doubly Robust Estimation and Inference. *ICML 2026*. ([Preprint](#))

Jain, S.*, Seale-Carlisle, T. M.*, Lee, C., Levenson, C., Ramprasad, S., Garrett, B., Roy, S., Rudin, C., & Volfovsky, A. (2024). Evaluating pre-trial programs using machine learning matching algorithms. *AAAI '24 (oral)*. ([Publication](#))

Kornfein, C., **Jain, S.***, Willard, F.*, Tang, C.*, Long, Y.*, Malof, J.*, Ren, S.*, & Bradbury, K.* (2022). Closing the Domain Gap - Blended Synthetic Imagery for Climate Object Detection. *NeurIPS '22 (Tackling Climate Change with Machine Learning Workshop)*. ([Publication](#)). Extended: *Environmental Data Science*. (2023). ([Publication](#))

Nayak, G. K., Mopuri, K. R., **Jain, S.**, & Chakraborty, A. (2021). Mining Data Impressions from Deep Models as Substitute for Unavailable Training Data. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. ([Publication](#))

Jain, S., Seth, G., Paruthi, A., Soni, U., & Kumar, G. (2020). Synthetic data augmentation for surface defect detection and classification using deep learning. *Journal of Intelligent Manufacturing*, 1-14. ([Publication](#))

Nayak, G. K., **Jain, S.**, Babu, R. V., & Chakraborty, A. (2020). Fusion of Deep and Non-Deep Methods for Fast Super-Resolution of Satellite Images. *IEEE International Conference on Multimedia Big Data (BigMM)* (pp. 267-271). ([Publication](#))

Jain, S., **Jain, S.**, & Komanduri, A. P. (2018). A cascade predictive control strategy for active suspension systems. *IEEE International Conference on Automation/XXIII Congress of the Chilean Association of Automatic Control (ICA-ACCA)* (pp. 1-5). ([Publication](#))

PROFESSIONAL & RESEARCH EXPERIENCE

JUN 2025 - SEP 2025 TOKYO, JAPAN	Research Intern , RIKEN AIP & University of Tokyo (Host: Masaaki Imaizumi) Developing novel high-dimensional contextual bandit algorithms using precise training dynamics for <i>finite-width</i> deep neural networks.
JUN 2024 - SEP 2024 NYC, USA	Applied Scientist II Intern , Amazon Science - Finetuned Large Language Models (LLMs) within deep sequence models for multi-horizon time-series forecasting. - Integrated embeddings of text data using pre-trained sentence transformers to improve forecasting performance for diverse product categories.
JUL 2023 - SEP 2023 LA, USA	AI/ML Engineering Intern , involve.ai - Built a research-and-revise approach for post-hoc LLM output correction, implemented as a 'co-pilot' for market research tasks. - Built a DistilBERT multi-task classifier for job titles, improving recall for sales lead retrieval by 4x.
AUG 2021 - MAY 2022 NC, USA	Graduate Team Lead , Bass Connections Project, Duke Energy Initiative Developed a synthetic imagery-based domain adaptation technique for energy infrastructure detection, improving cross-domain AP by 8.23%.
MAY 2021 - APR 2022	Graduate Research Assistant , Almost Matching Exactly Lab, Duke University

NC, USA	Developed an interpretable ML-based approach for sub-population level impact evaluation of judicial pre-trial services on recidivism.
MAY 2021 - AUG 2021 NC, USA	Research Intern , RAILabs, Duke University Developed an end-to-end YOLO-based pipeline for lesion detection and joint ipsilateral matching in mammograms, reducing false positives by 80% with minimal speed impact.
JAN 2020 - JUL 2020 KARNATAKA, INDIA	Research Project Assistant , Visual Computing Lab, IISc Bangalore - Developed a novel statistical approach for constructing training data 'impressions' from pre-trained CNN parameters, advancing the SOTA in several data-free ML tasks. - Created a model-agnostic technique for fast selective superresolution, matching deep model performance at near-bicubic interpolation speed.
JUN 2019 - JAN 2020 UP, INDIA	Machine Learning Engineer , Million Sparks Foundation - Developed an unsupervised domain adaptation network for real-time user outcome prediction, achieving ~4.3% improvement over SOTA AUC scores - Built models for text classification using BERT, recommendation using neural collaborative filtering, and explainable (with SHAP) churn prediction.

SELECT PROJECTS

Out-of-Distribution (OOD) Detection using Likelihood Ratios ([Project](#))

- Implemented likelihood ratio-based approaches that construct a background-contrastive statistic, to correct for spurious background information that can confound deep generative models into assigning high likelihood to OOD image inputs.

Window Level Optimization for Pectoral Muscle Segmentation in MLO-view Mammograms ([Project](#))

- Formulated the window-levelling operation (the typically deterministically-set function for mapping a 12–16 bits/pixel grayscale DICOM image to an 8-bit grayscale display) as a convolution layer.
- Jointly optimized it with a U-Net, to obtain an improvement of ~4.5% on pectoral muscle segmentation.

Bachelor's Thesis: Synthetic Data Augmentation for Surface Defect Detection and Classification using Deep Learning ([Project](#))

- Proposed a [highly-cited](#) synthetic data augmentation approach for (typically small) manufacturing defect datasets to offset the high opportunity cost of data acquisition, surpassing the classic augmentation performance by ~5.5%.

TECHNICAL SKILLS

Languages: Python, R, C/C++, Matlab

Frameworks/Libraries: PyTorch, TensorFlow, Keras, OpenCV

Technologies: AWS, Apache Airflow, GCP, Git, Flask, REST, FastAPI

AWARDS & RECOGNITION

MAY 2026	WNAR 2026 Outstanding Written Student Paper Award (Runner Up)
MAY 2026	NESS 2026 Student Research Award
APR 2026	Graduate Conference Presentation Award by UW, for presenting at ACIC 2026
SEP 2022	Coursera Department Fellowship by UW, for outstanding promise for graduate work
AUG 2021	CVSPK Talent Incentive Scheme by NSIT, award for outstanding undergraduate research
JUL 2019	Merit Award , computer vision and machine learning summer schools at IIIT Hyderabad for scoring in the top-20 out of 280 participants
MAY 2019	Bachelor's Thesis Ranked 2 in the Dept. of Manufacturing Processes and Automation Engg.
JUN 2018	International Rank 20 , CanSat Competition 2018

TEACHING EXPERIENCE (PRE-DOC TEACHING ASSOCIATE)

AUTUMN 2024	Visualizing Data (STAT 451), Dept. of Statistics, UW Seattle
SPRING 2023	Introduction to Statistical Machine Learning (STAT 435), Dept. of Statistics, UW Seattle
WINTER 2023	Statistical Methods in Engineering and Science (STAT 390), Dept. of Statistics, UW Seattle
FALL 2022	Elements of Statistical Methods (STAT 311), Dept. of Statistics, UW Seattle
SPRING 2022	Probabilistic Machine Learning (STA 561D), Dept. of Statistics, Duke University
FALL 2021	Vector Space Methods with Applications (ECE 586), Dept. of ECE, Duke University